

Is IT A Prime?

Input file: `standard input`
Output file: `standard output`
Time limit: 1 second
Memory limit: 256 megabytes

Given an integer n . your task is simply to check it if is **prime or not***

Input

given an integer n ($2 \leq n \leq 1000$).

Output

Print YES if the given number is prime. otherwise print NO

Examples

standard input	standard output
3	YES
8	NO

*A prime number is a natural number greater than 1 and has exactly 2 divisors which are 1 and the number itself.